1.43 Trigonometry: Modelling in Degrees

Chapter 9 · Original slide 44 · Sin Graph

90 180 270 360

What does it look like?

Key Features:

• Repeats every 360°.

• Range: $-1 < \sin(x) < 1$

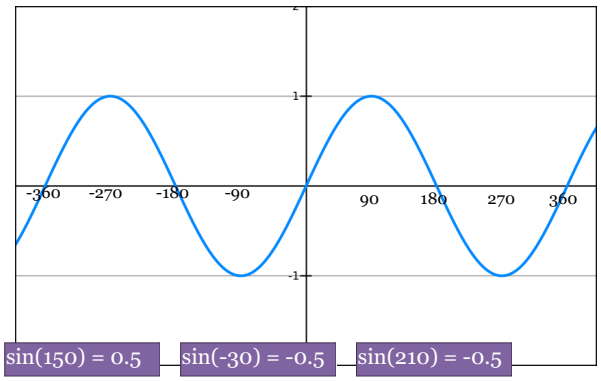
-360

-270

-180

-90

Chapter 9 · Original slide 45 · Sin Graph



What do the following graphs look like?

Suppose we know that sin(30) = 0.5. By thinking about symmetry in the graph, how could we work out:

1.43 Trigonometry: Modelling in Degrees

Chapter 9 · Original slide 46 · Cos Graph

90 180 270 360

What do the following graphs look like?

Key Features:

- Repeats every 360°.
- Range: $-1 < \sin(x) < 1$

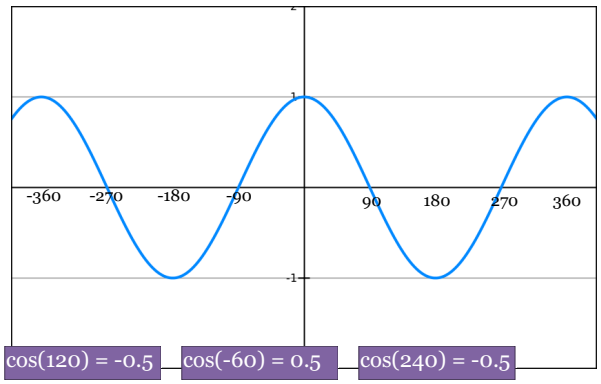
-360

-270

-180

-90

Chapter 9 · Original slide 47 · Cos Graph



What does it look like?

Suppose we know that cos(60) = 0.5. By thinking about symmetry in the graph, how could we work out:

1.43 Trigonometry: Modelling in Degrees

Chapter 9 · Original slide 48 · Tan Graph

90 180 270 360

What does it look like?

-360

-270

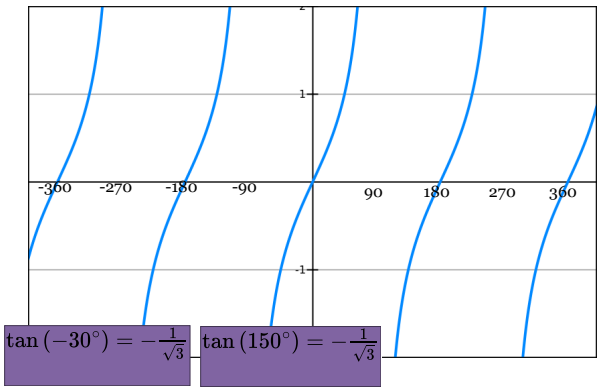
-180

-90

Key Features:

- Repeats every $\mathbf{180}^\circ$.
- Roots: $0, 180^\circ, -180^\circ, \ldots$
- Range: $\tan(x) \in \mathbb{R}$ (i.e. no min/max value!)
- Asymptotes: $x = \pm 90^\circ, \pm 270^\circ, \ldots$

Chapter 9 · Original slide 49 · Tan Graph

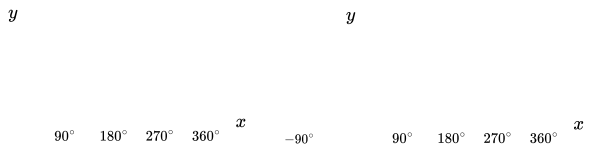


What does it look like?

Suppose we know that $\tan(30^\circ) = \frac{1}{\sqrt{3}}$. By thinking about symmetry in the graph, how could we work out:

1.43 Trigonometry: Modelling in Degrees

Chapter 9 · Original slide 50 · Transforming Trigonometric Graphs

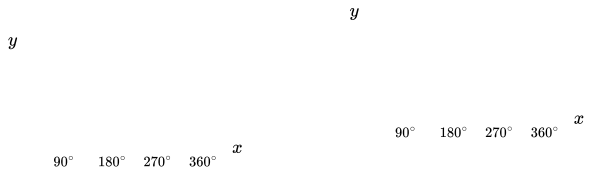


Sketch $y = 4\sin x,\ 0 \leq x \leq 360^\circ$

There is no new theory here: just use your knowledge of transforming graphs, i.e. whether the transformation occurs 'inside' the function (i.e. input modified) or 'outside' the function (i.e. output modified).

Sketch $y = \cos(x + 45^\circ),\ 0 \leq x \leq 360^\circ$

Chapter 9 · Original slide 51 · Transforming Trigonometric Graphs



Sketch $y = \sin\left(\frac{x}{2}\right),\ 0 \leq x \leq 360^\circ$

Sketch $y = -\tan x,\ 0 \leq x \leq 360^\circ$

1.43 Trigonometry: Modelling in Degrees

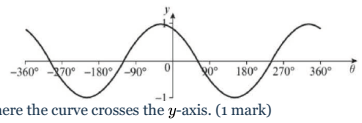
Chapter 9 · Original slide 52 · Exercise 9F – Y1 Page 194

1 Sketch the graph of $y = \cos\theta$ in the interval $-180^\circ \leq \theta \leq 180^\circ$.
2 Sketch the graph of $y = \tan\theta$ in the interval $-180^\circ \leq \theta \leq 180^\circ$.
3 Sketch the graph of $y = \sin\theta$ in the interval $-90^\circ \leq \theta \leq 270^\circ$.

Chapter 9 · Original slide 53 · Exercise 9G – Y1 Pages 197-198

The graph shows the curve $y = \cos(x + 30°)$, $-360° \leqslant x \leqslant 360°$.

a Write down the coordinates of the points where the curve crosses the x-axis. **(2 marks)**



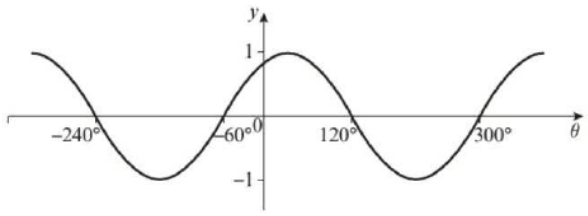
b Find the coordinates of the point where the curve crosses the y-axis. (1 mark)

7 The graph shows the curve $y = \cos(x + 30^\circ), -360^\circ \leq x \leq 360^\circ$.
a Write down the coordinates of the points where the curve crosses the x-axis. (2 marks)

b $\cos 30^\circ = \dfrac{\sqrt{3}}{2}$; $\left(0, \dfrac{\sqrt{3}}{2}\right)$

1.43 Trigonometry: Modelling in Degrees

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8 The graph shows the curve with equation $y = \sin(x + k)$, $-360^\circ \leq x \leq 360^\circ$, where k is a constant.
a Find one possible value for k. (2 marks)
b Is there more than one possible answer to part a? Give a reason for your answer. (2 marks)

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9 The variation in the depth of water in a rock pool can be modelled using the function $y = \sin(30t)^\circ$, where t is the time in hours and $0 \leq t \leq 6$.
a Sketch the function for the given interval.
b If $t = 0$ represents midday, during what times will the rock pool be at least half full?